

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2025

Course: B. Tech.

Branch: Civil Engineering

Semester: VIII

Subject Code & Name: (BTCESS802E) Soil Structures Interaction

Max Marks: 60

Date:13/ 01/ 2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	The main limitation of Terzaghi's bearing capacity theory is that it	CO1	1				
	<table border="1"> <tr> <td>A. ignores effect of water table</td> <td>B. assumes soil to be elastic</td> <td>C. neglects settlement criteria</td> <td>D. considers only cohesive soils</td> </tr> </table>	A. ignores effect of water table	B. assumes soil to be elastic	C. neglects settlement criteria	D. considers only cohesive soils		
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2	Net bearing capacity is obtained by	CO1	1				
	<table border="1"> <tr> <td>A. subtracting surcharge from ultimate capacity</td> <td>B. adding surcharge to safe bearing capacity</td> <td>C. multiplying ultimate capacity by factor of safety</td> <td>D. subtracting self-weight of footing only</td> </tr> </table>	A. subtracting surcharge from ultimate capacity	B. adding surcharge to safe bearing capacity	C. multiplying ultimate capacity by factor of safety	D. subtracting self-weight of footing only		
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3	Soil–structure interaction mainly affects	CO1	1				
	<table border="1"> <tr> <td>A. only bearing capacity</td> <td>B. only settlement</td> <td>C. contact pressure distribution</td> <td>D. depth of foundation only</td> </tr> </table>	A. only bearing capacity	B. only settlement	C. contact pressure distribution	D. depth of foundation only		
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4	Two-parameter models improve Winkler model by considering	CO2	1				
	<table border="1"> <tr> <td>A. soil weight</td> <td>B. continuity of soil</td> <td>C. footing rigidity</td> <td>D. pore water pressure</td> </tr> </table>	A. soil weight	B. continuity of soil	C. footing rigidity	D. pore water pressure		
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5	Infinite beam assumption is valid when beam length is	CO2	1				

	A. equal to footing width	B. much smaller than span	C. very large compared to characteristic length	D. zero		
6	Simply supported beam on elastic foundation has				CO2	1
	A. zero deflection at supports	B. zero slope at supports	C. zero moment at mid span	D. uniform soil pressure		
7	Layered soil response differs from homogeneous soil due to				CO3	1
	A. constant stiffness	B. varying modulus	C. uniform pressure	D. rigid base		
8	Two-parameter foundation consider				CO3	1
	A. compression only	B. shear only	C. compression and shear	D. damping only		
9	Plate on elastic foundation is mainly used for				CO4	1
	A. strip footings	B. raft foundations	C. pile caps	D. retaining walls		
10	Computer programs for soil–structure interaction mainly solve				CO4	1
	A. algebraic equations	B. differential equations	C. empirical relations	D. graphical methods		
11	Group efficiency of piles may be				CO5	1
	A. always greater than one	B. always equal to one	C. less than one	D. independent of spacing		
12	Uplift capacity of pile is governed by				CO5	1
	A. end bearing	B. skin friction	C. bending resistance	D. pile weight only		
Q. 2	Solve the following.					12
A)	Explain the effect of rise in water table on bearing capacity and settlement of footing on sand.				CO1	6

B)	What are the factors affecting bearing capacity of shallow foundations?	CO1	6
Q.3	Solve the following.		12
A)	Explain foundation models used to idealize soil for SSI problems.	CO2	6
B)	Explain the concept of beam on elastic foundation.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Discuss time-dependent response of soil.	CO3	6
B)	What determines stiffness of a beam? Explain relative stiffness and absolute stiffness.	CO3	6
C)	Explain significance of beam end conditions on deflection and bending moment behaviour.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain Poisson–Kirchhoff plate theory for thin plates.	CO4	6
B)	Explain application of FDM in solving one-dimensional consolidation equation.	CO4	6
C)	How does soil-structure interaction affect structural response and its applications?	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Explain classification of piles based on load transfer mechanism.	CO5	6
B)	Explain significance of pile load testing in verification of pile capacity.	CO5	6
C)	Describe Reese and Matlock method for laterally loaded piles with assumptions.	CO5	6
	*** End ***		